



NEW YORK AIRBNB 2019 REVENUE OPTIMIZATION

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NYC AIRBNB

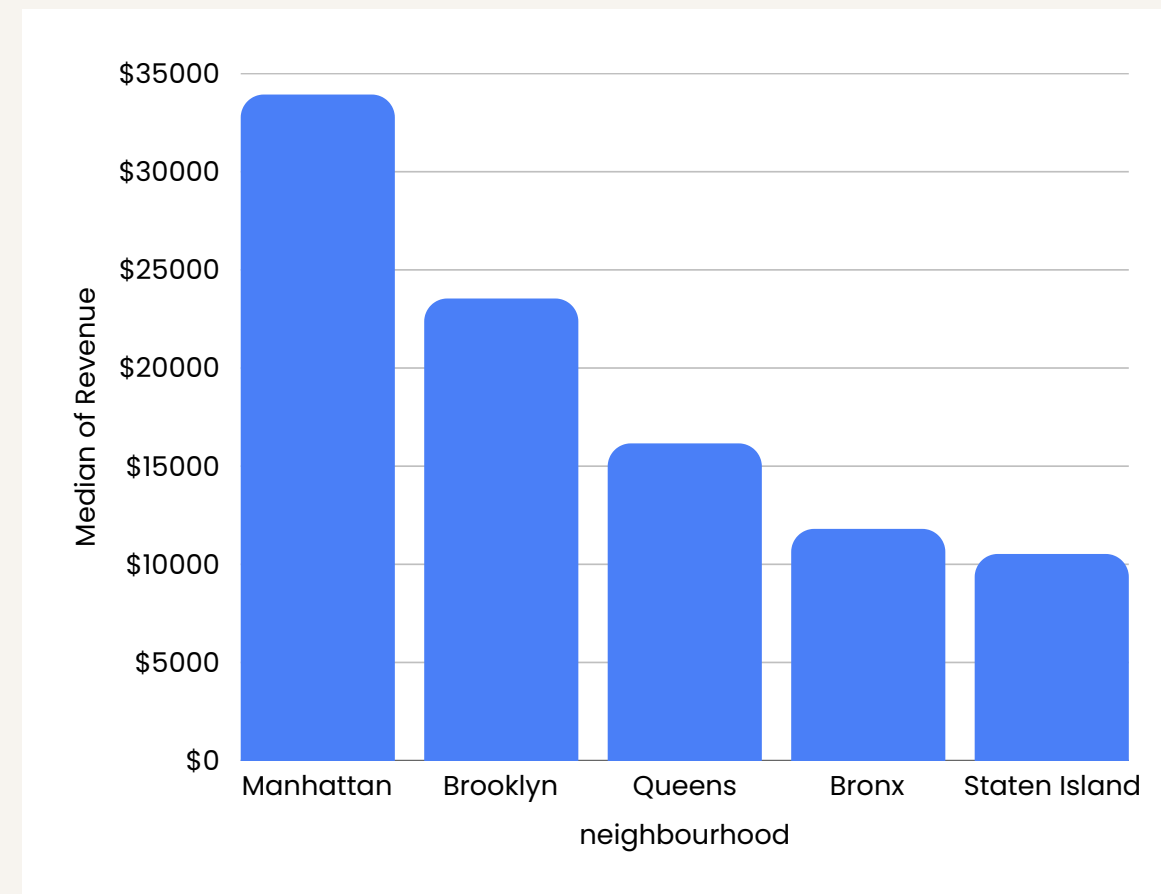
With nearly 49,000 Airbnb listings distributed across 221 neighbourhoods and five boroughs, New York City represents one of Airbnb's largest and most competitive markets.

The diversity of locations and property types creates significant variation in revenue performance, making it increasingly important for hosts to understand where the most profitable opportunities exist. By analyzing key revenue drivers such as location, pricing, occupancy, and room type, hosts can make more informed decisions to maximize revenue and improve market positioning



MAIN CHALLENGE FOR NYC AIRBNB

REVENUE OPTIMIZATION IN COMPETITIVE MARKET



How can AirBnB hosts can maximize revenue across different boroughs and room types?



THE PROBLEM



Should Manhattan pricing and room type **strategy** should be applied to other borough?

What is the best room type to **increase revenue** for each borough?



THE GOAL

To decide what is the **best revenue driver** for room type for each borough for owners and investors



ANALYTICAL APPROACH

Data Collection & Preprocessing

- Browsing the Airbnb data which provide required elements.
- Performing data cleaning and creating necessary calculation.

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- Data retrieved from Kaggle “airbnb nyc 2019”
- Anomaly Handling
 - Price = 0 => Remove
- Creating new data calculation
 - Estimated revenue
 - Occupancy Rate
 - Revenue per available room
 - occupied days
- Finalizing with 48,879 entries with 20 columns

Exploratory Data Analysis (EDA)

- Identifying key factors contributing to Revenue per available room.
- Conducting statistical analysis to examine the data relevancy

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- RevPar derived from $((\text{price} * \text{occupied days}) * (\text{occupancy rate}))$
- Calculating Median for RevPar and Price for each borough and validating with kruskal-wallis to verify huge discrepancy among boroughs.
- Analyzing median RevPar for boroughs and room type to show the discrepancy between them.
- Using Power BI Shape Map, showing which area has more RevPar among the other

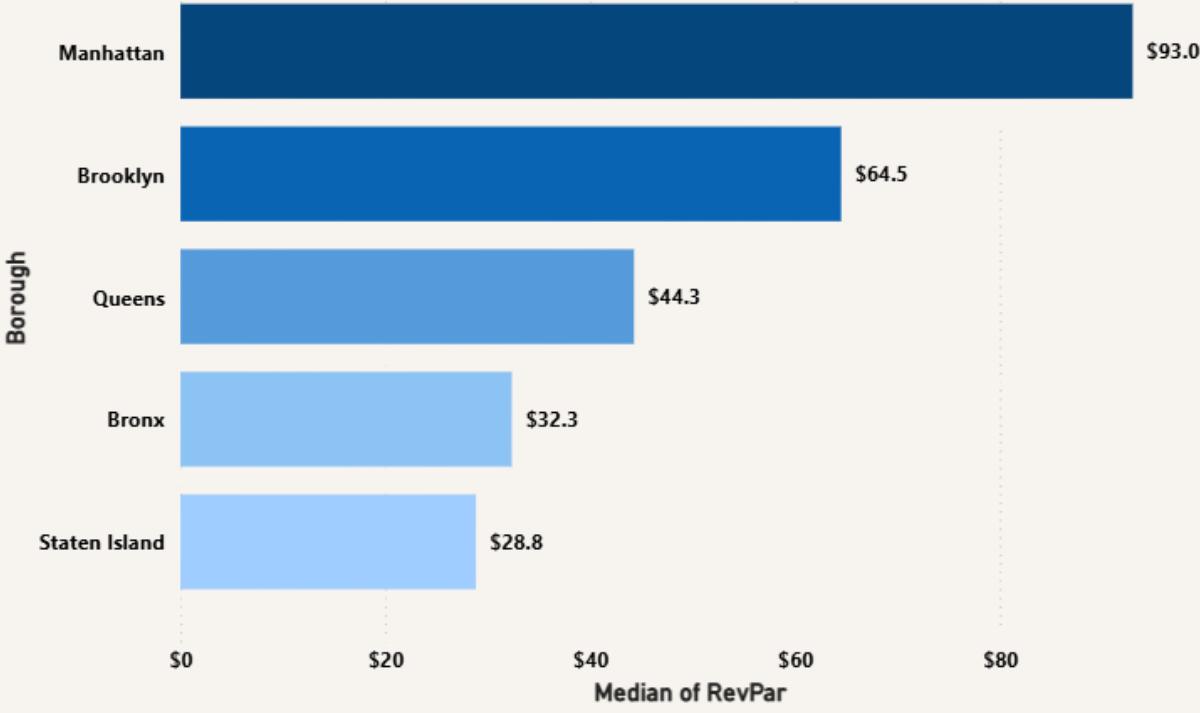
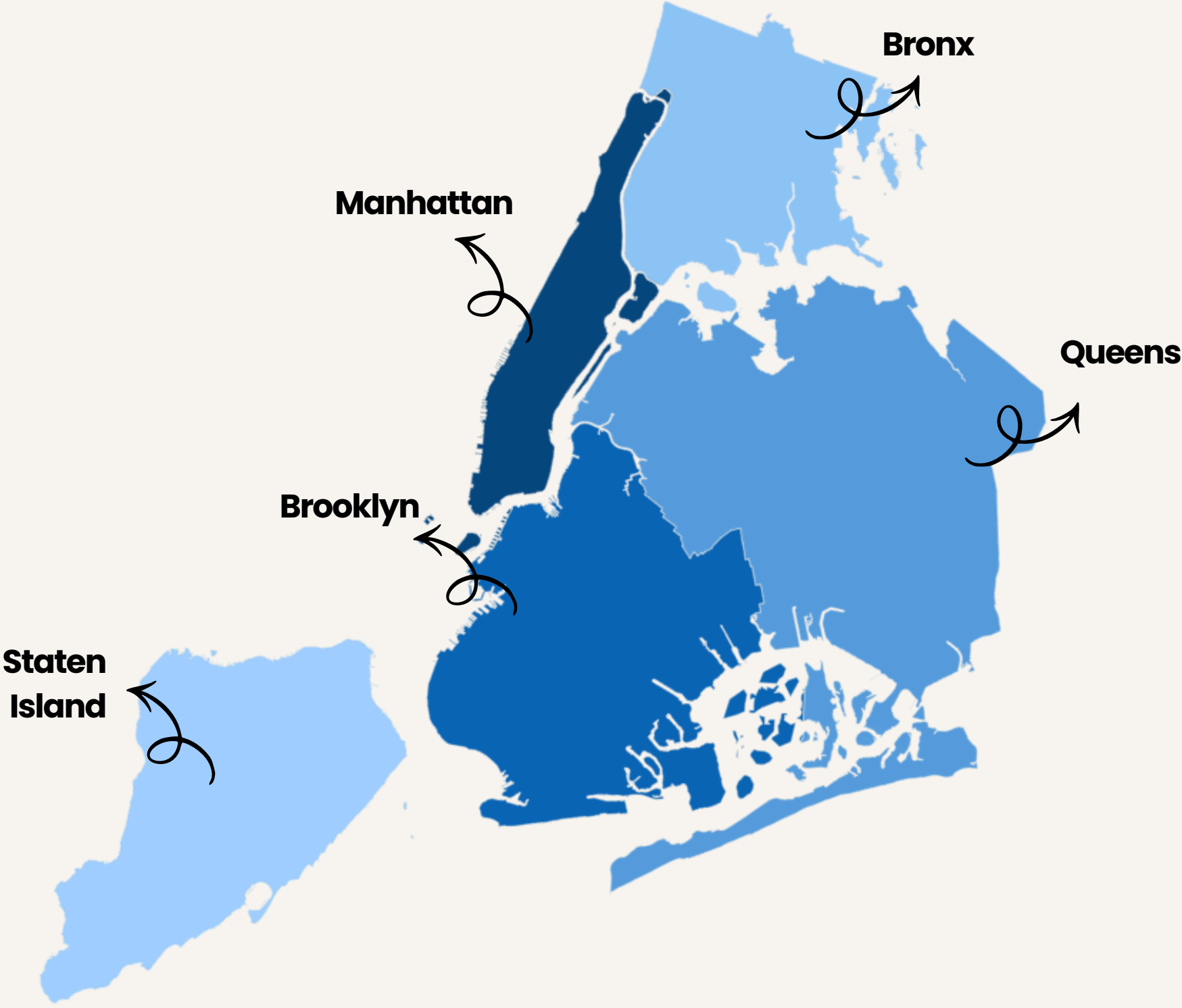


DATA UNDERSTANDING

ID	=	Listing identification (unique number)
Name	=	Airbnb listing name
minimum_night	=	minimum book/spend night
Room_type	=	Type of the listing
Neighbourhood_group	=	Borough
Availability_365	=	How many days the unit is vacant, or unoccupied
Occupied_days	=	Booked days
Occupancy_rate	=	Percentage of booked days
Revenue	=	Price X booked days
RevPar	=	Average revenue earned per available room considering both price and occupancy



REVPAR BY BOROUGH

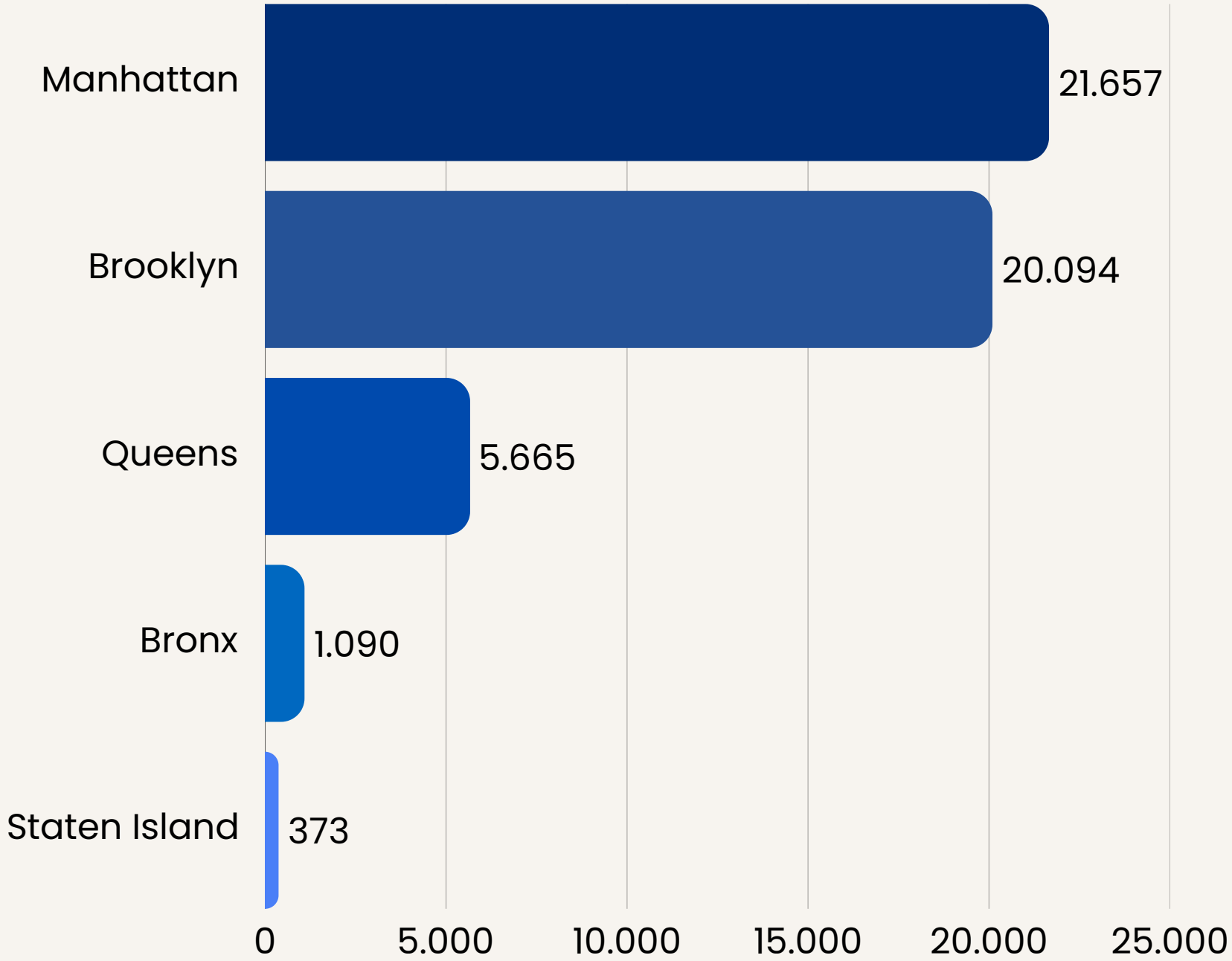


Manhattan generates the highest median RevPAR (\$93) despite being one of NYC's smallest boroughs, driven by strong tourism and business demand.

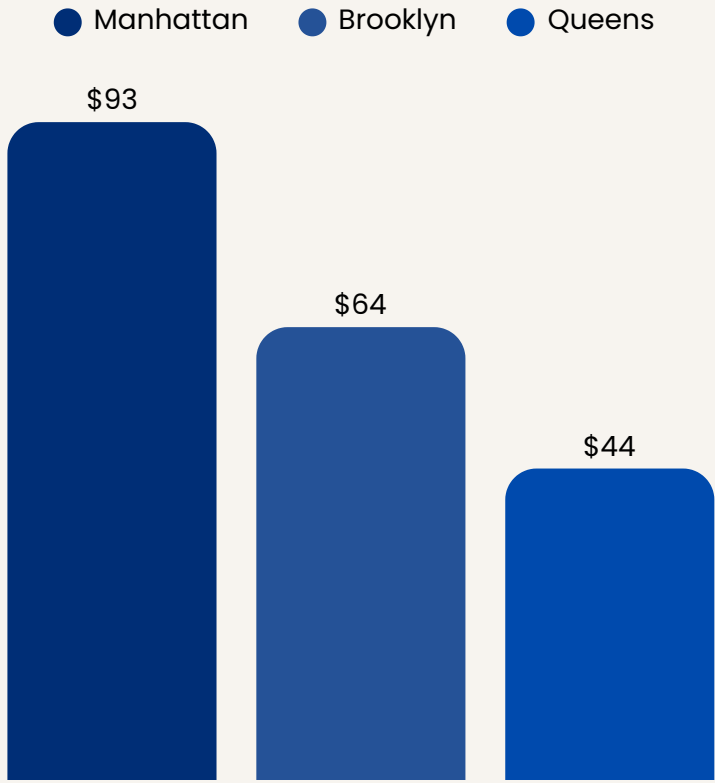
In contrast, Staten Island records the lowest RevPAR (\$28.8), reflecting lower demand and accessibility. This highlights that location attractiveness has a greater impact on Airbnb revenue than geographic size.



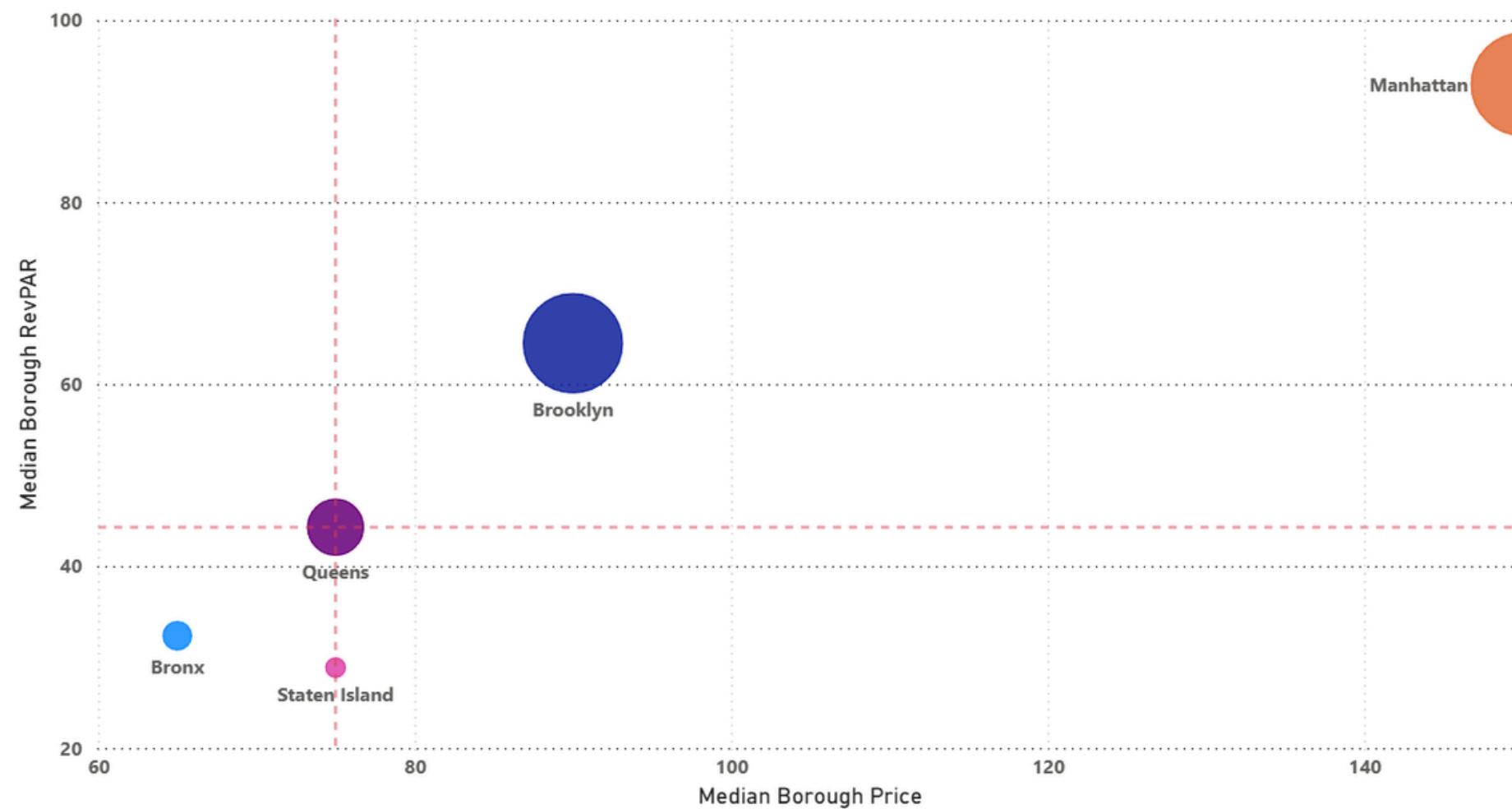
BOROUGH LISTING COUNT



One of the key factor is listing count, with Manhattan being highest listing, but it does not stop on it, price plays significant role on the RevPar contribution, Manhattan and brooklyn have only slight different in listing count, but queens have huge different, means there is other factor play a significant role to the RevPar contribution



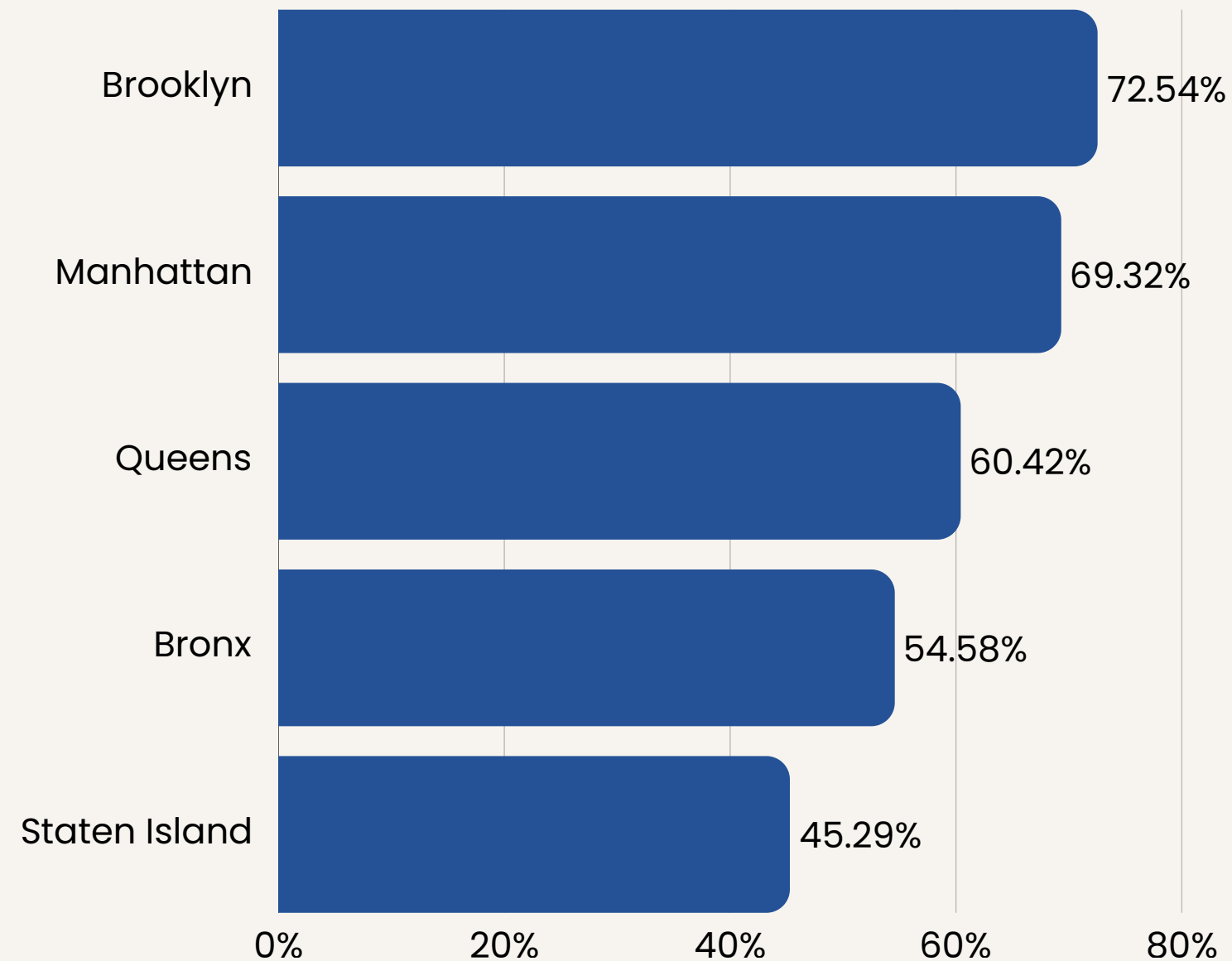
REVENUE OPPORTUNITY MATRIX



- **Manhattan and Brooklyn** are in the high performance market, with high RevPar and Price
- **Queens** stayed in average market performance
- **Staten Island and Bronx** is in the underperforming market with low price and low RevPar



OCCUPANCY RATE BY BOROUGH



- **Brooklyn** has the highest demand, higher than Manhattan
- **Staten Island** has the hardest challenge on fulfilling the demand
- Most NYC Airbnb listings are booked for more than half of the year, highlighting strong market demand and attractive revenue opportunities for both new and existing hosts

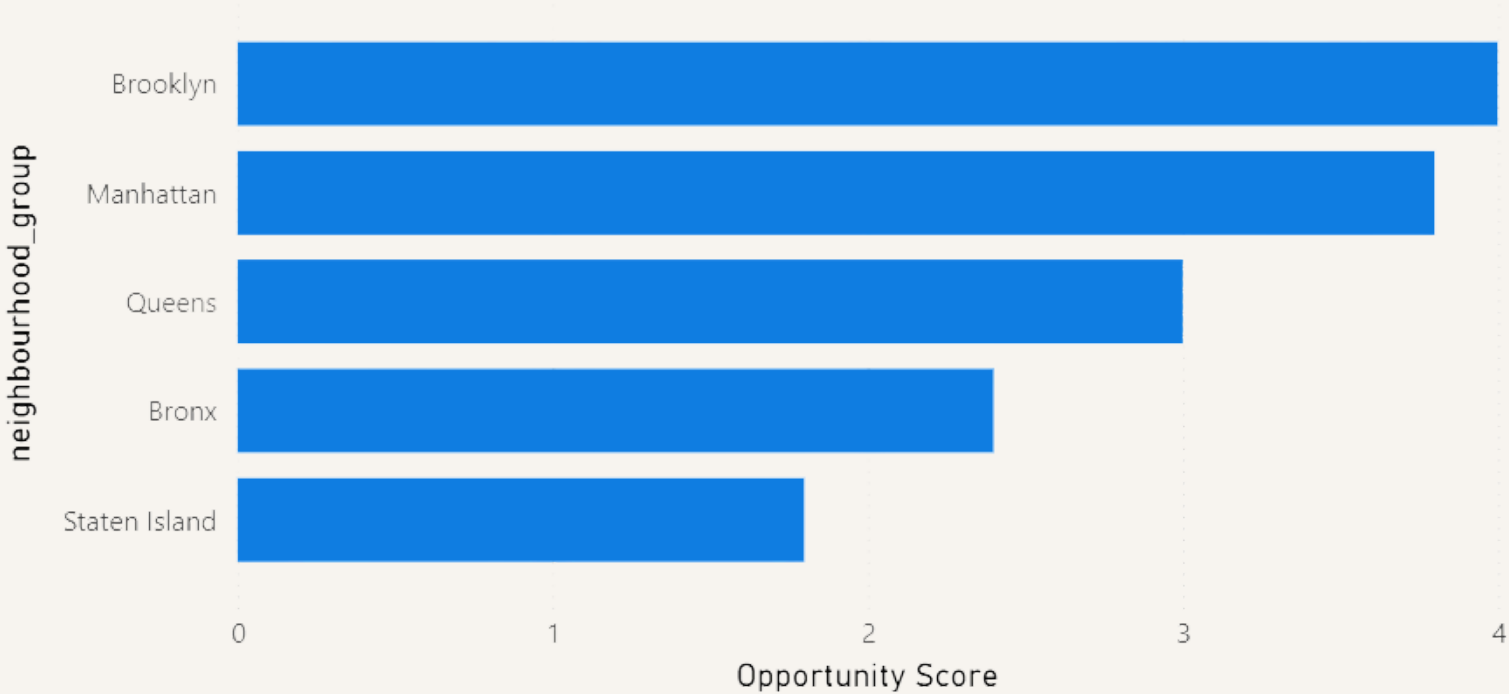


BOROUGH RECOMMENDATION

Borough	RevPar	Occupancy	Listing Count	Score
Brooklyn	64.50	72.54%	20094	4.00
Manhattan	92.96	69.32%	21657	3.80
Queens	44.27	60.42%	5665	3.00
Bronx	32.34	54.58%	1090	2.40
Staten Island	28.84	45.29%	373	1.80

The proportion of scoring is 40%, RevPar, 40% Occupancy, and 20% competitiveness, generate the best RevPar driver borough is **Brooklyn**.

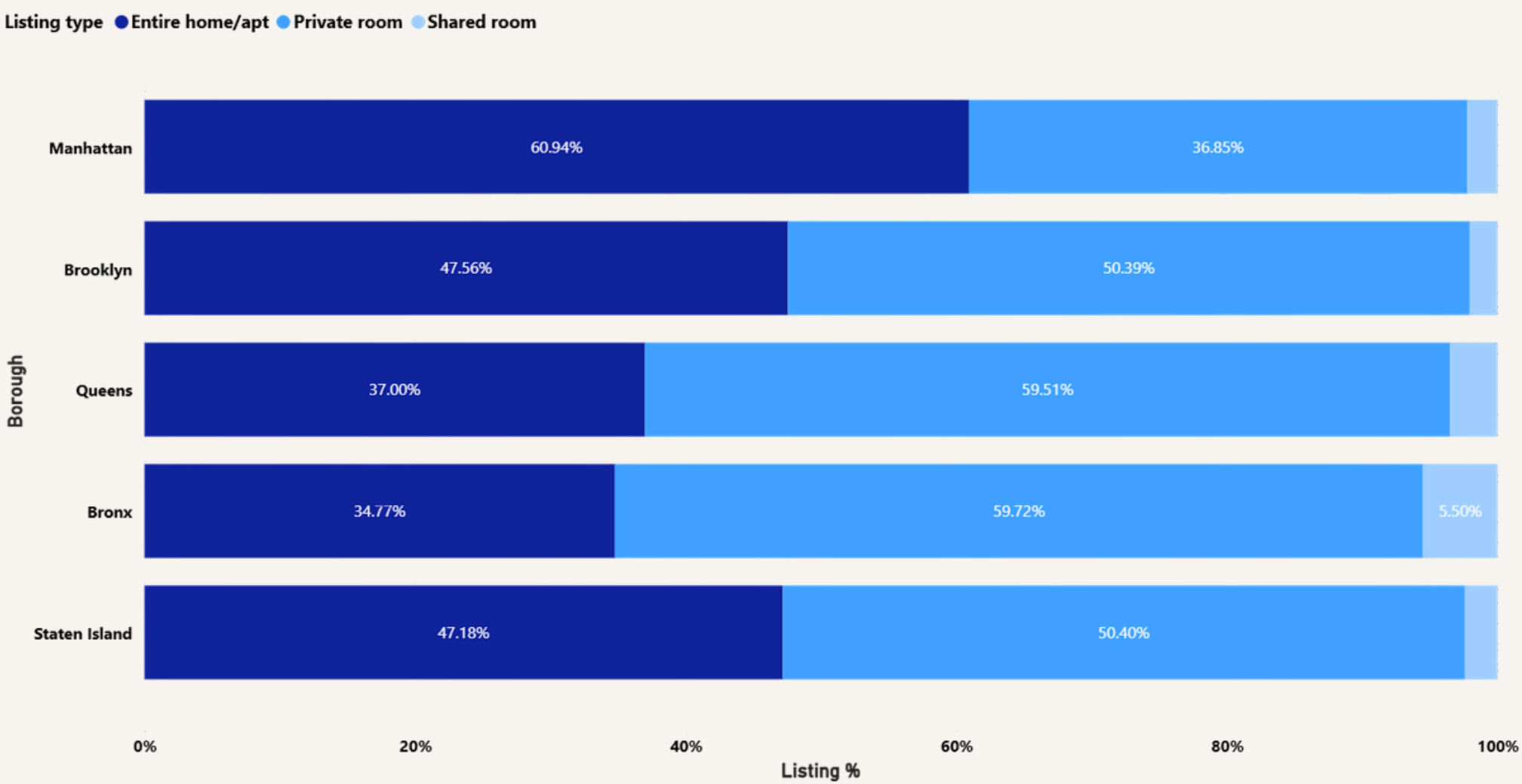
Opportunity Score by neighbourhood_group



- **Brooklyn** offers the best balance of demand and RevPar generation.
- **Manhattan** remains the premium revenue market
- **Queens** presents a growth opportunity, with moderate RevPar and occupancy rate, potential for cost conscious investors



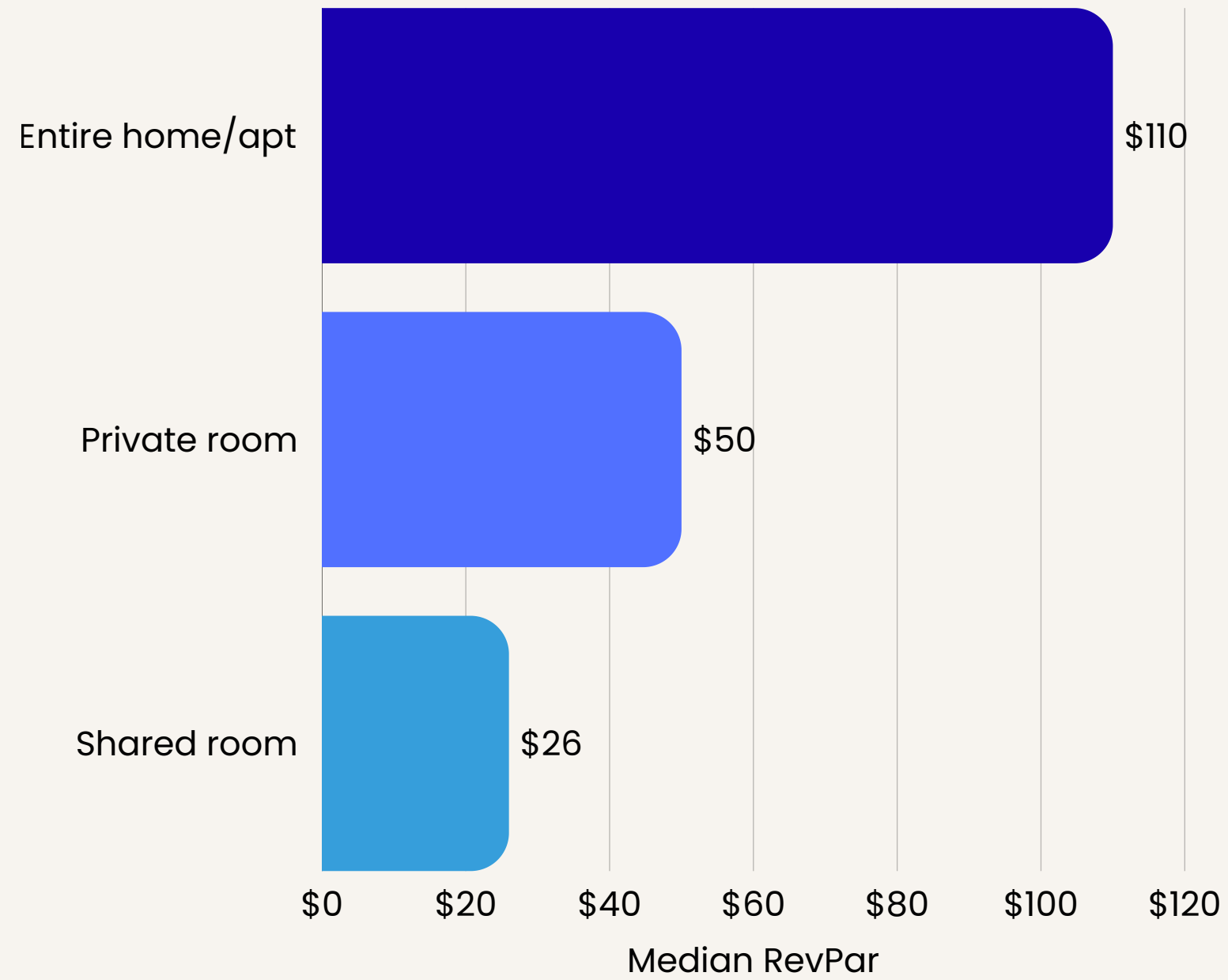
LISTING TYPE LISTING PER BOROUGH



- **Manhattan** is driven by Entire Home listings, while **Queens and Bronx** are primarily supported by Private Rooms, indicating both of listing type is the most preferred by guests.
- **Shared Rooms** representing less than 6% of listings in every borough, revenue optimization efforts should primarily focus on Entire Homes and Private Rooms.



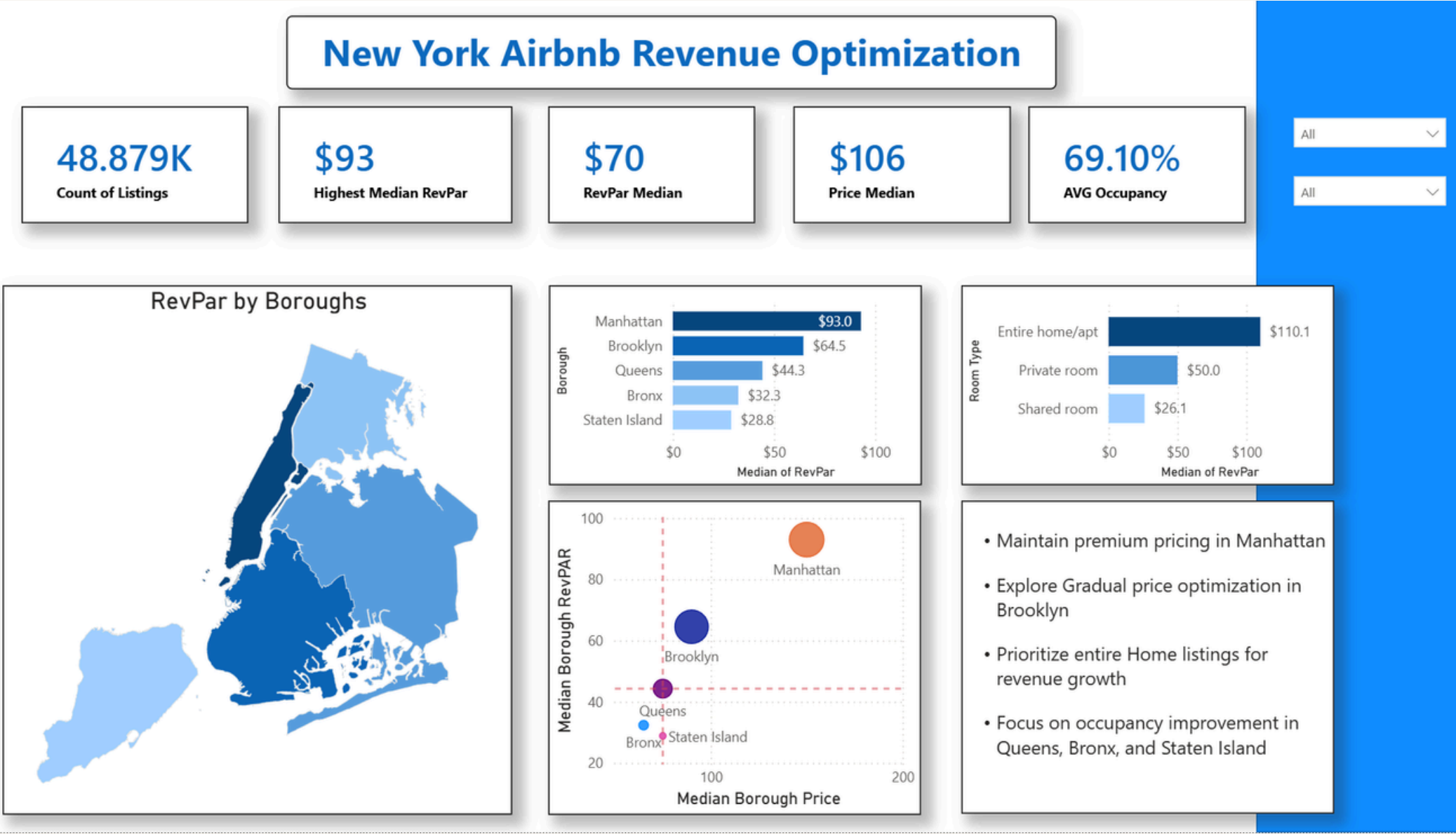
REVPAR BY ROOM TYPE



Entire Home significantly outperforms other room types in generating revenue



POWER BI DASHBOARD





THANKYOU

